

Lab Assignment 1:

Evaluation of AI-Generated Images

Objective

Evaluate the quality, diversity, and text alignment of AI-generated images using Inception Score, Fréchet Inception Distance, and CLIP Score.

Dataset

1. AGIQA 3K: Use 1,000 AI-generated images with their corresponding prompts.
(<https://github.com/lcysyzxdc/AGIQA-3k-Database>)
2. KoniQ 10k: Use 1,000 real images as the reference set for FID.
(<https://darus.uni-stuttgart.de/dataset.xhtml?persistentId=doi%3A10.18419%2Fdarus-2435>)

Tasks

1. Load the 1,000 AI-generated images from AGIQA 3K.
2. Load the 1,000 real images from KoniQ 10k.
3. Compute the Inception Score for the AI-generated images.
4. Compute the FID between the KoniQ real image set and the AGIQA generated image set.
5. Compute the CLIP Score between every AGIQA image and its corresponding text prompt.
6. Report the mean and standard deviation of the CLIP Scores.
7. Compare the results and explain:
 - a. What does a higher Inception Score indicate?
 - b. What does a lower FID indicate?
 - c. What does a higher CLIP Score indicate?
 - d. Why are IS, FID and CLIP Score measuring different properties?
 - e. What are the limitations of computing FID between KoniQ 10k and AGIQA 3K?

Evaluation of Generated Text

Objective

Evaluate generated text against reference text using BLEU, ROUGE and BERTScore, and analyze the differences between lexical and semantic evaluation metrics.

Dataset

Use the provided CSV file:

text_evaluation_1000.csv

Tasks

1. Load the CSV file using Pandas.
2. Preprocess the reference and candidate texts.
3. Compute the following metrics:
 - BLEU 1
 - BLEU 2
 - BLEU 3
 - BLEU 4
 - ROUGE 1
 - ROUGE 2
 - ROUGE L
 - BERTScore
4. Report the average score for each metric over all 1,000 samples.
5. Compare the results across different topics.
6. Identify examples where:
 - a. BLEU is low, but BERTScore is high.
 - b. ROUGE is low, but BERTScore is high.
 - c. BLEU is relatively high despite limited semantic quality.
 - d. Different metrics produce different conclusions.
7. Explain:
 - a. Why does BLEU depend strongly on n-gram overlap?
 - b. How is ROUGE different from BLEU?
 - c. Why can BERTScore identify semantic similarity despite different wording?
 - d. Why should no single metric be used as the only measure of text quality?

